8. Hywind, the world's first floating wind turbine that combines technologies from both the wind farming industry and the oil and gas sectors was recently commissioned off the coast of Norway in 2009. The Hywind are towed out to the open sea of density 1025 kg m^{-3} and is anchored to the seabed by three long cables as shown in Fig. 8.1. The high-speed wind turbine is used to generate electricity of output frequency of 50 Hz. One of the reasons for operating the Hywind offshore is the continuous presence of high speed wind from 8.0 m s^{-1} to 30 m s^{-1}.

Fig. 8.2 shows the different parts inside the generator housing.

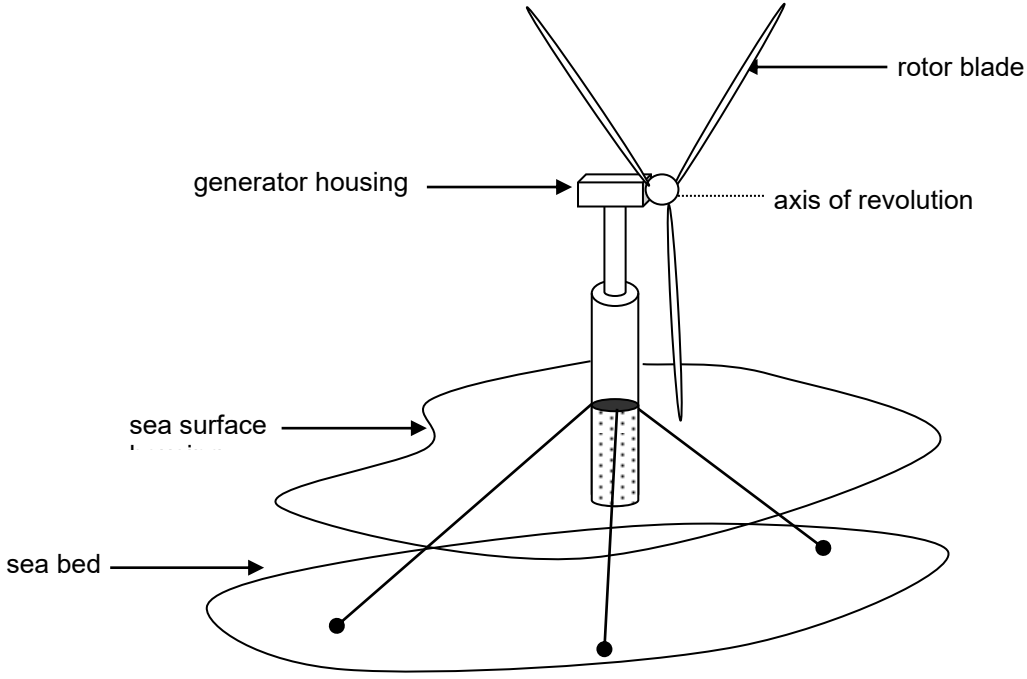

Fig. 8.1

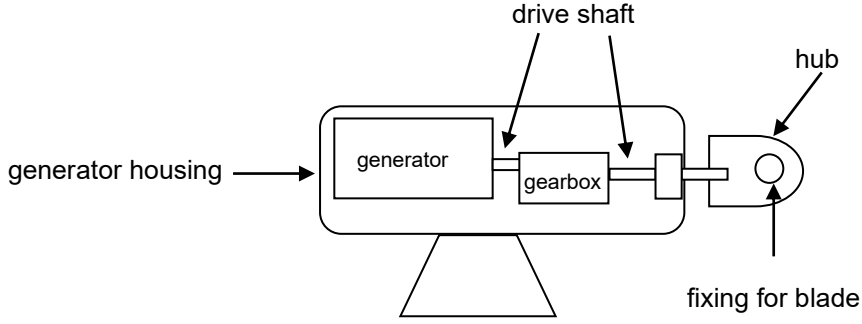

Fig. 8.2

Some information provided by the manufacturer of Hywind is given in Fig. 8.3.

Turbine mass	138 000 kg
Turbine height above the sea surface	65.0 m
Rotor blade diameter	82.4 m
Displacement of water by Hywind	5300 m^3
Diameter of submerged body	8.3 m
Range of water depth	120 - 700 m
Frequency	50 Hz

Fig. 8.3

(a) State and explain whether the generator produces direct current or alternating current.

.....

.....

..... [1]

(b) Determine the upthrust on the Hywind structure.

upthrust = ... N [1]

(c) **(i)** The turbine is located so that it faces the wind. The rotor blades are set at an angle to the plane in which they revolve so that the wind is deflected.

Explain why the rotor blades are able to revolve about the axis when subjected to the wind.

.....

.....

..... [3]

(ii) Describe briefly how electrical energy is generated using the energy from the wind.

.....

.....

.....

..... [2]

Question 8 continues on next page

(d) Fig. 8.4 shows how the output power, *P*, from a wind turbine varies with the speed, *v*, of the wind.

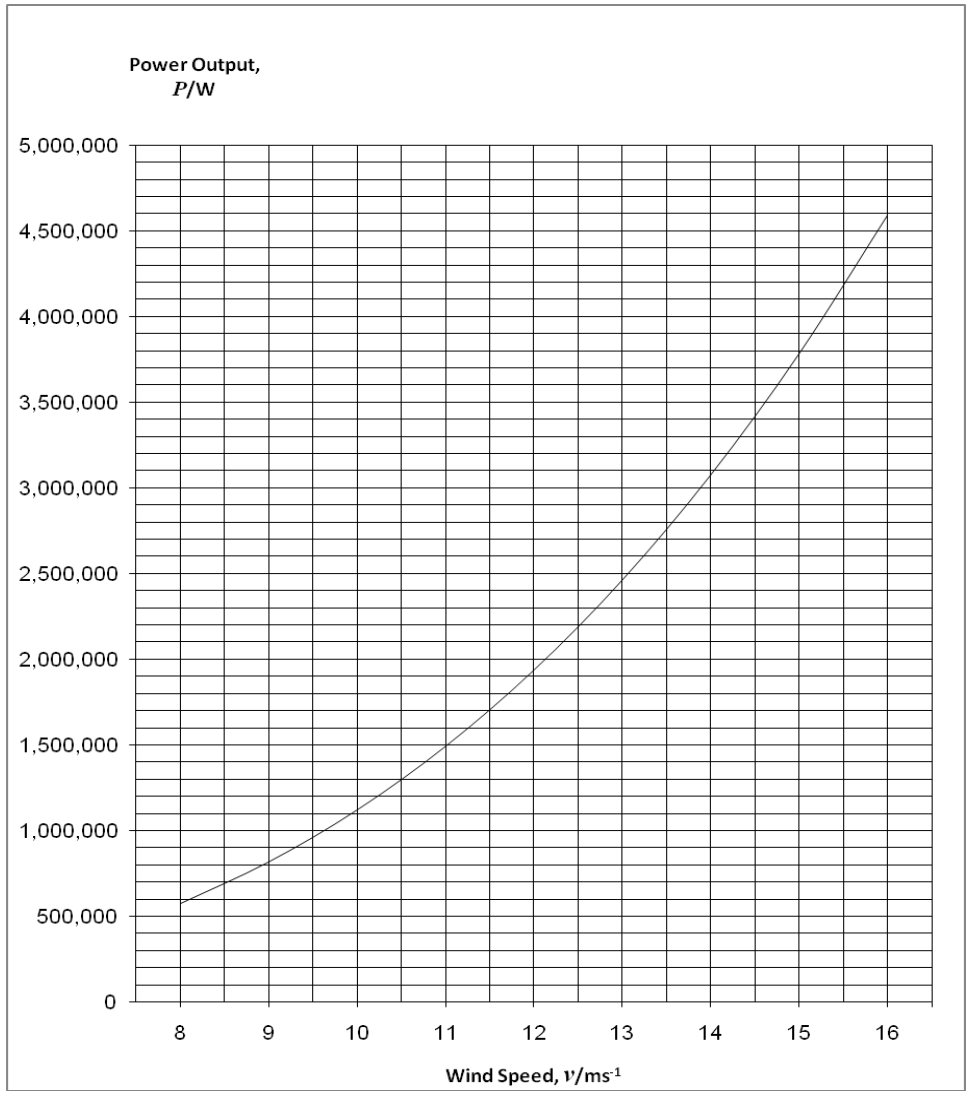

Fig. 8.4

It is thought that, for a given fixed size of the rotor blade, the electrical power output, *P*, varies with the wind speed *v* according to the expression

$$P = kv^3$$

Using the graph of Fig. 8.4, show that *k* is a constant.

[2]

(e) Some corresponding values of lg P and lg v for the data in Fig. 8.4 are plotted on the graph of Fig. 8.5.

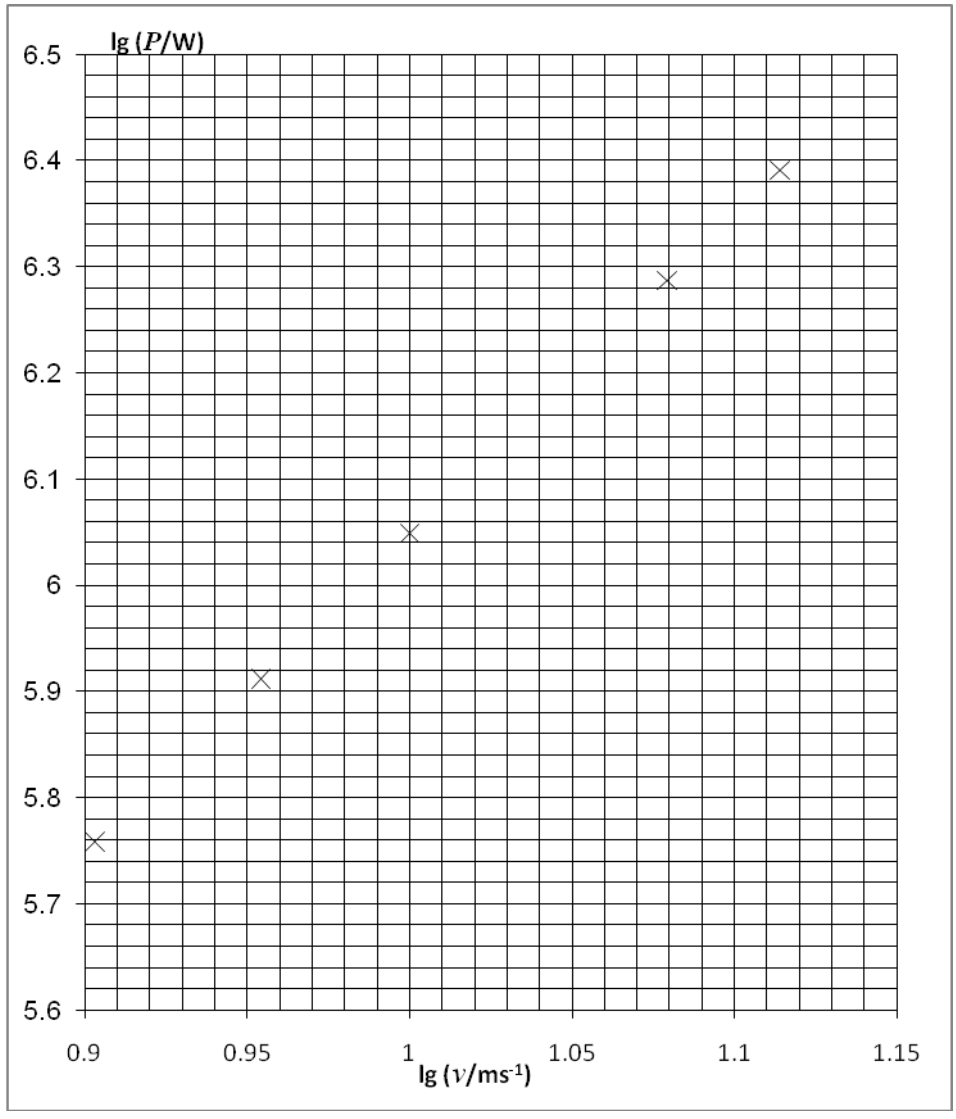

Fig. 8.5

(i) On Fig. 8.5,

1. plot the point corresponding to v = 11.0 m s^{-1}, [1]

2. draw the best fit line for all the plotted points. [1]

(ii) Determine the gradient of the line drawn in **(i) part 2**.

gradient = ... [2]

(iii) Hence comment on the validity of the relation given in **part (d)**.

Explain your answer.

.....

.....

.....

..... [2]

(f) The initial kinetic energy per second of the air that passes through the wind turbine is given by $\frac{1}{2}\rho A v^3$, where ρ is the density of air and A is the area swept out by the rotor blades in each rotation.

Given: density of air = 1.2 kg m^{-3}

Calculate, for the wind turbine operating at wind speed of 11.0 m s^{-1},

(i) kinetic energy of air incident per second on the rotor blades,

incident wind power = ... MW [2]

(ii) the overall efficiency of generation of electric power.

efficiency = ... [1]

(g) Suggest two possible problems encountered when operating the Hywind in the open sea.

1. .....

.....

2. .....

..... [2]